Nervous System

Lab 01 – Nervous tissue

Models:

- Multipolar neuron model
- Neuromuscular junction
- Synapse

Slides:

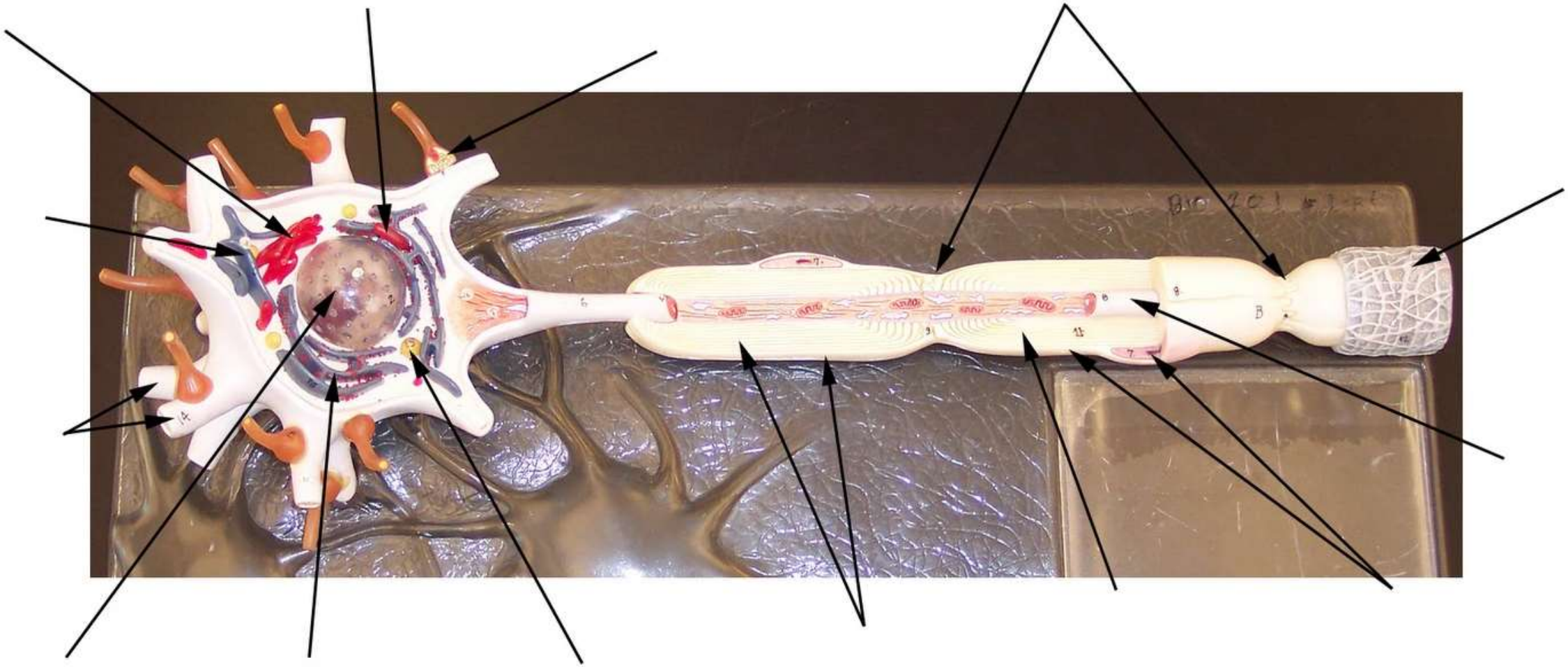
- Multipolar neuron with neuroglia
- Neuromuscular junction

List from page 9-9 of Lab Manual:
Learn Bold Terms!

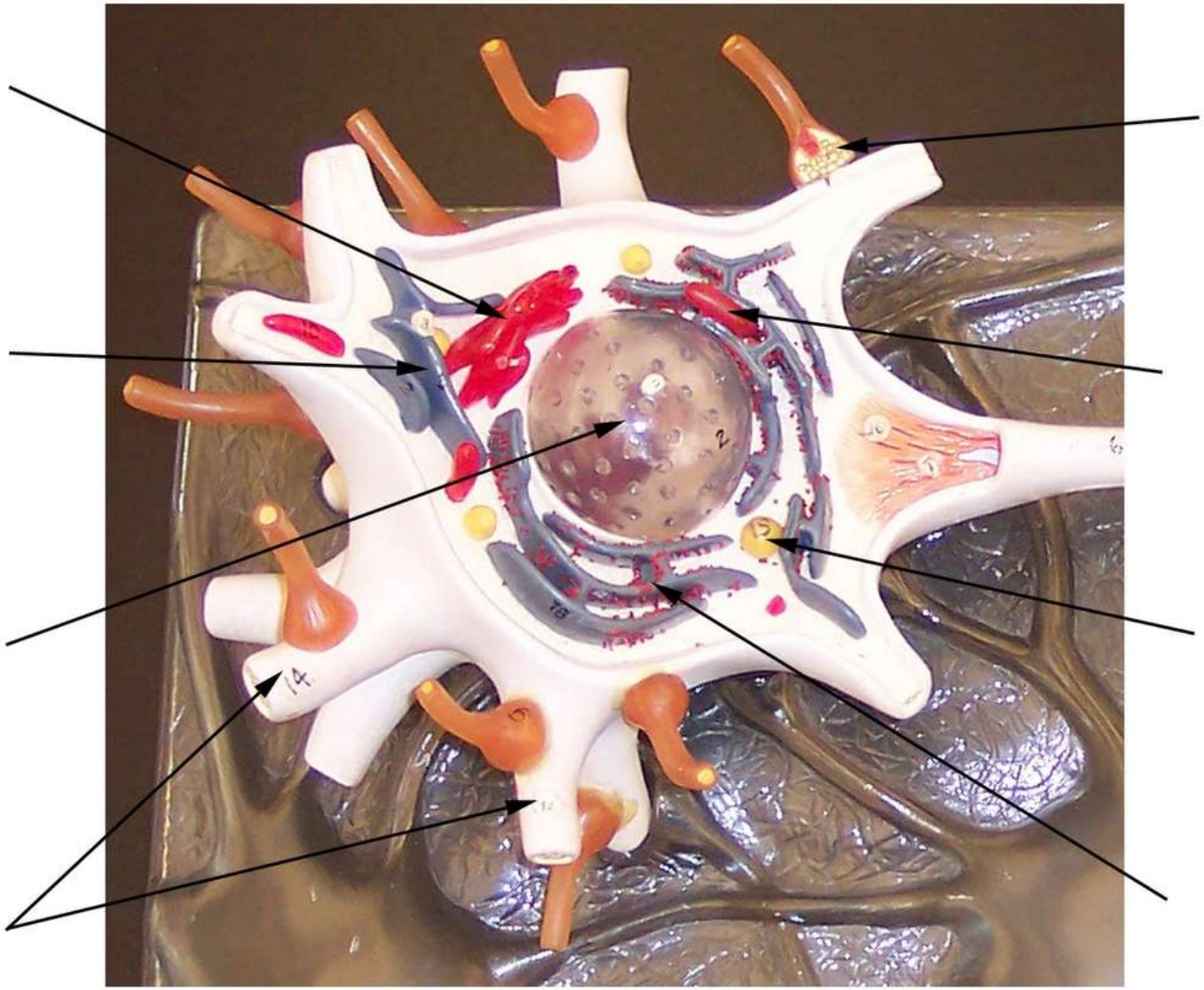
Multipolar Neuron Model

A	**Cell body**
B	**axon** with its **myelin sheath**
1	axon hillock, with abundant neurofibrils but lacking Nissl bodies
2	**nucleus** of neuron with a pair of nucleoli inside
3	**smooth endoplasmic reticulum**
5	**synaptic end bulbs** from presynaptic neurons
6	**axon**
7	**nucleus of Schwann cell**
8	**neurolemma**, the outer wrapping layer of the Schwann cell including its cell membrane, cytoplasm and nucleus
9	**node of Ranvier**
10	**mitochondria**
11	**myelin sheath**
12	**endoneurium**
14	**dendrites**
15	**lysosomes**
16	neurofibrils and microtubules
17	**golgi apparatus**
18	**Nissl bodies** = rough endoplasmic reticulum with its ribosomes.
19	**synaptic end bulbs** from presynaptic neurons

Multipolar Neuron model



Multipolar Neuron model: cell body region



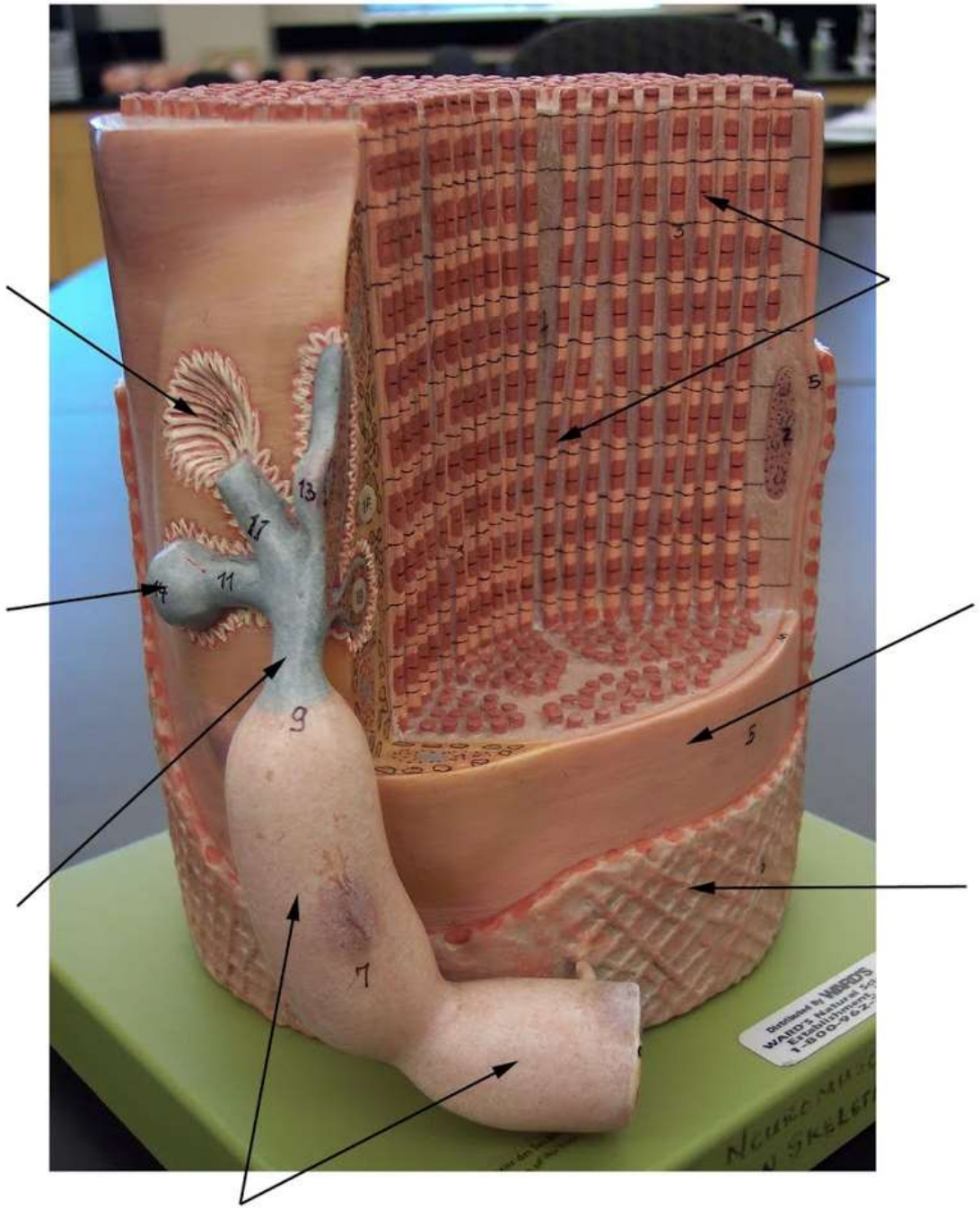
List from page 9-10 of Lab Manual:
Learn Bold Terms!

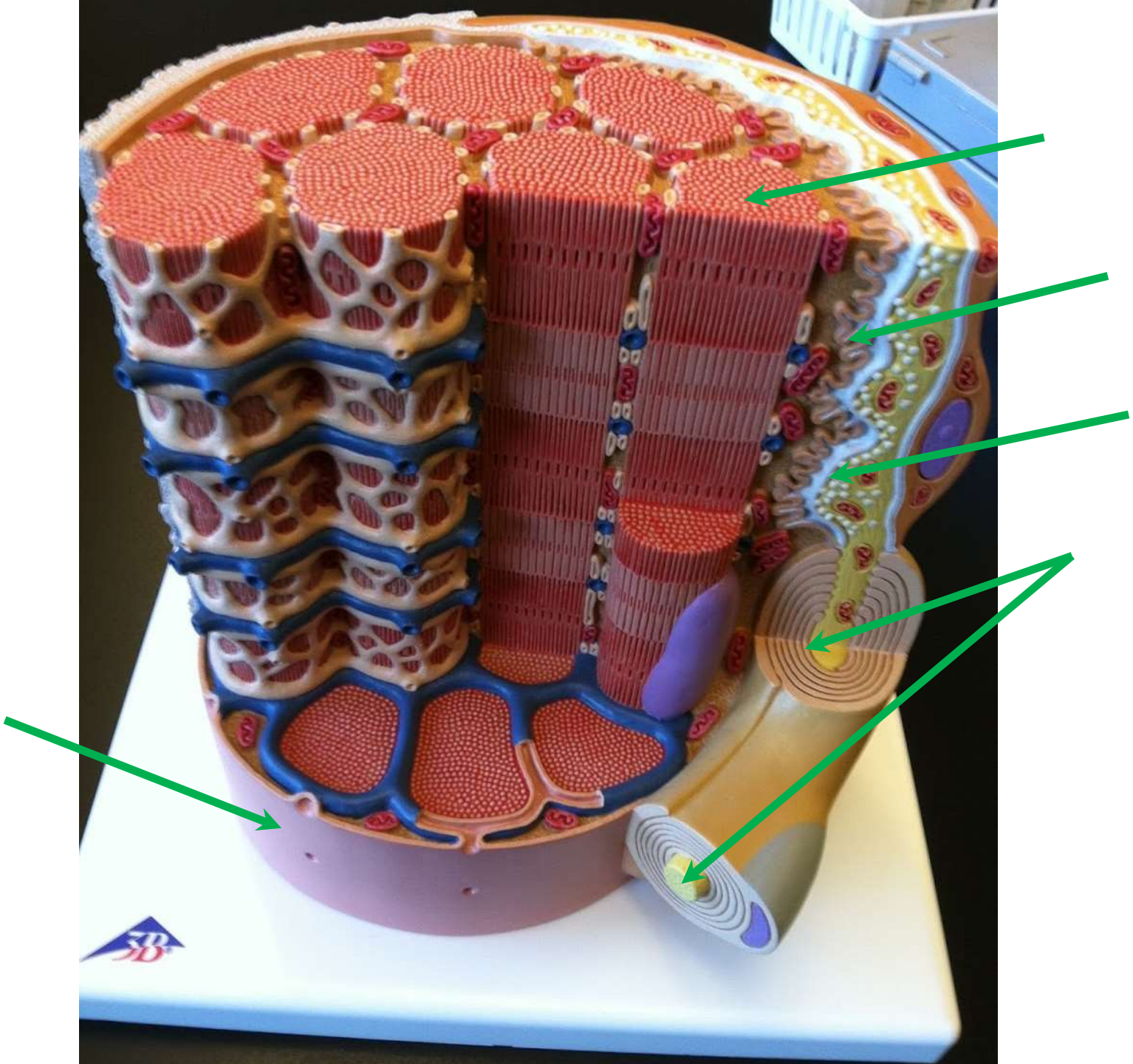
Skeletal (=Striated) Muscle Fiber with Neuromuscular Junction

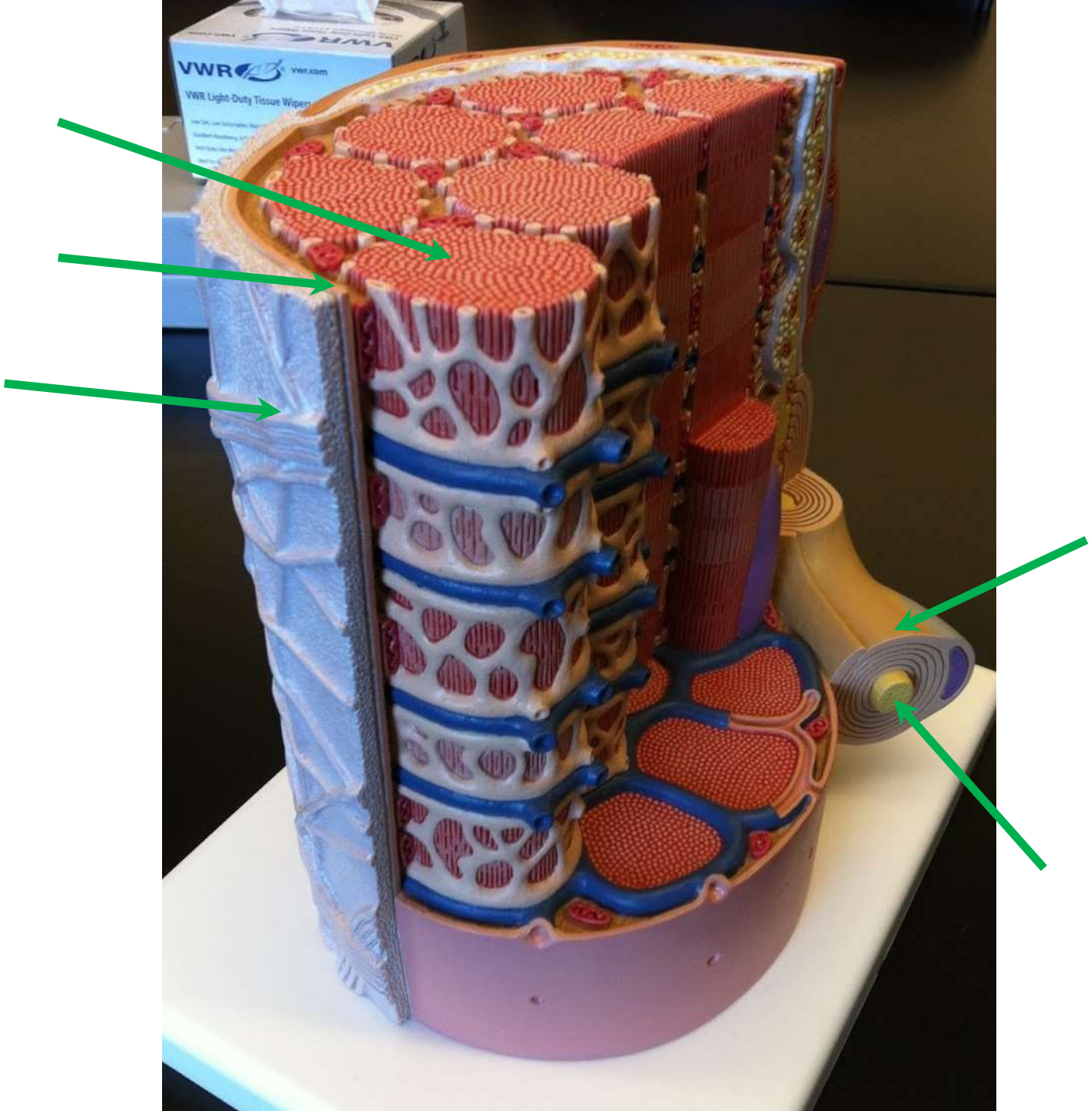
The model is approximately 4000x life size. At this magnification the detail of the sarcoplasmic reticulum and T-tubules present within the sarcoplasm in between the myofibrils, and the arrangement of the actin and myosin myofilaments within the myofibrils would not be revealed.

1 sarcoplasm
2 nuclei, peripherally located just deep to the sarcolemma.
3 **myofibrils**
5 **sarcolemma**, the muscle cell membrane.
6 **endomysium**
7 myelinated axon of a **motor neuron**
8 axon in cross section. The spirally layered wrappings of myelin (#8a) from the **Schwann cell** surround the axon.
10 & 19 **Synaptic end bulbs** sectioned.
17 & 18 The **motor end plate** is the specialized region of the sarcolemma adjacent to the axon terminals and synaptic end bulbs.

Neuromuscular Junction






VWR
vwr.com
VWR Light-Duty Tissue Wipers

List from page 9-9 of Lab Manual:
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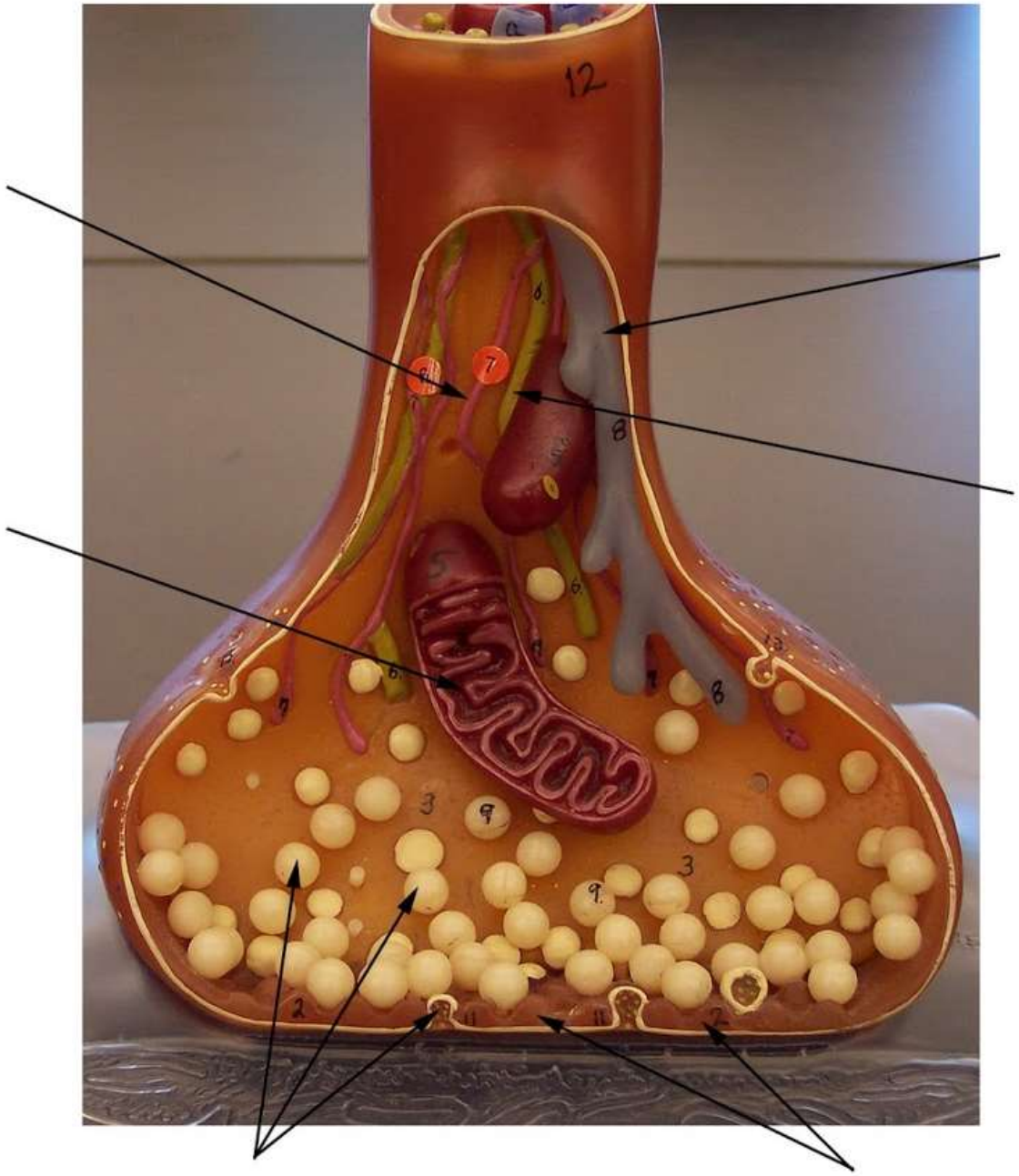
Model of a Synapse

The magnification of this model is approximately 120,000 times life size. The axon terminal of the presynaptic neuron is shown in detail. The clear plastic base represents the postsynaptic cell where structural details are not clearly shown. The synaptic cleft (=gap) is the space between the presynaptic and postsynaptic membranes. Compare the model to the diagrams in your textbook.

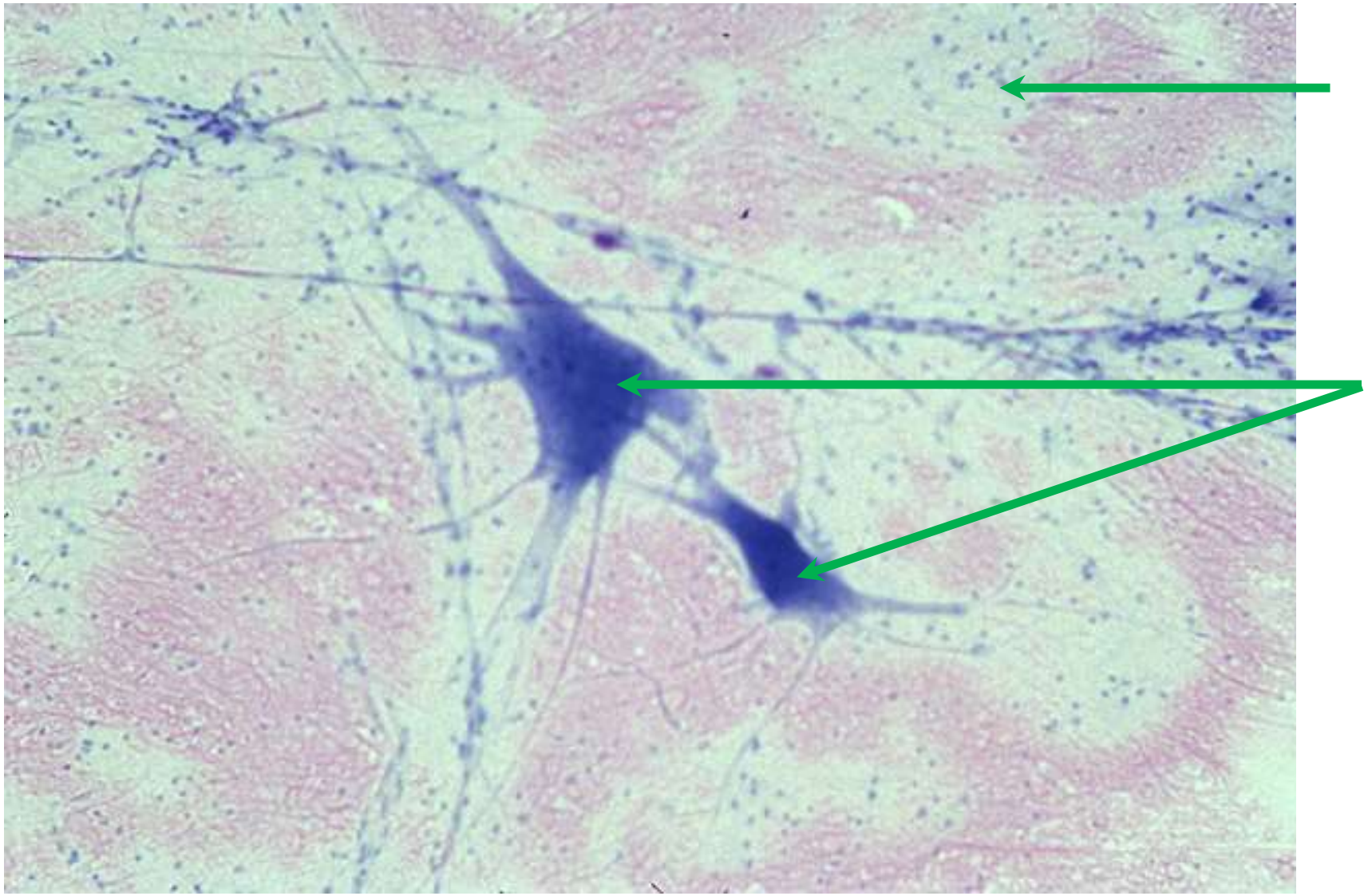
1 **synaptic end bulb** of an axon (= axon terminal bouton)
2 **presynaptic membrane**
5 **mitochondrium**
6 **microtubule**
7 **neurofibril**
8 **smooth endoplasmic reticulum**
9 **vesicles, containing neurotransmitter**
10 **vesicles** fusing with the presynaptic membrane prior to the release of neurotransmitter molecules

Synaptic End Bulb



Multipolar neuron slide

Description of this slide on page 9-2 of the lab manual



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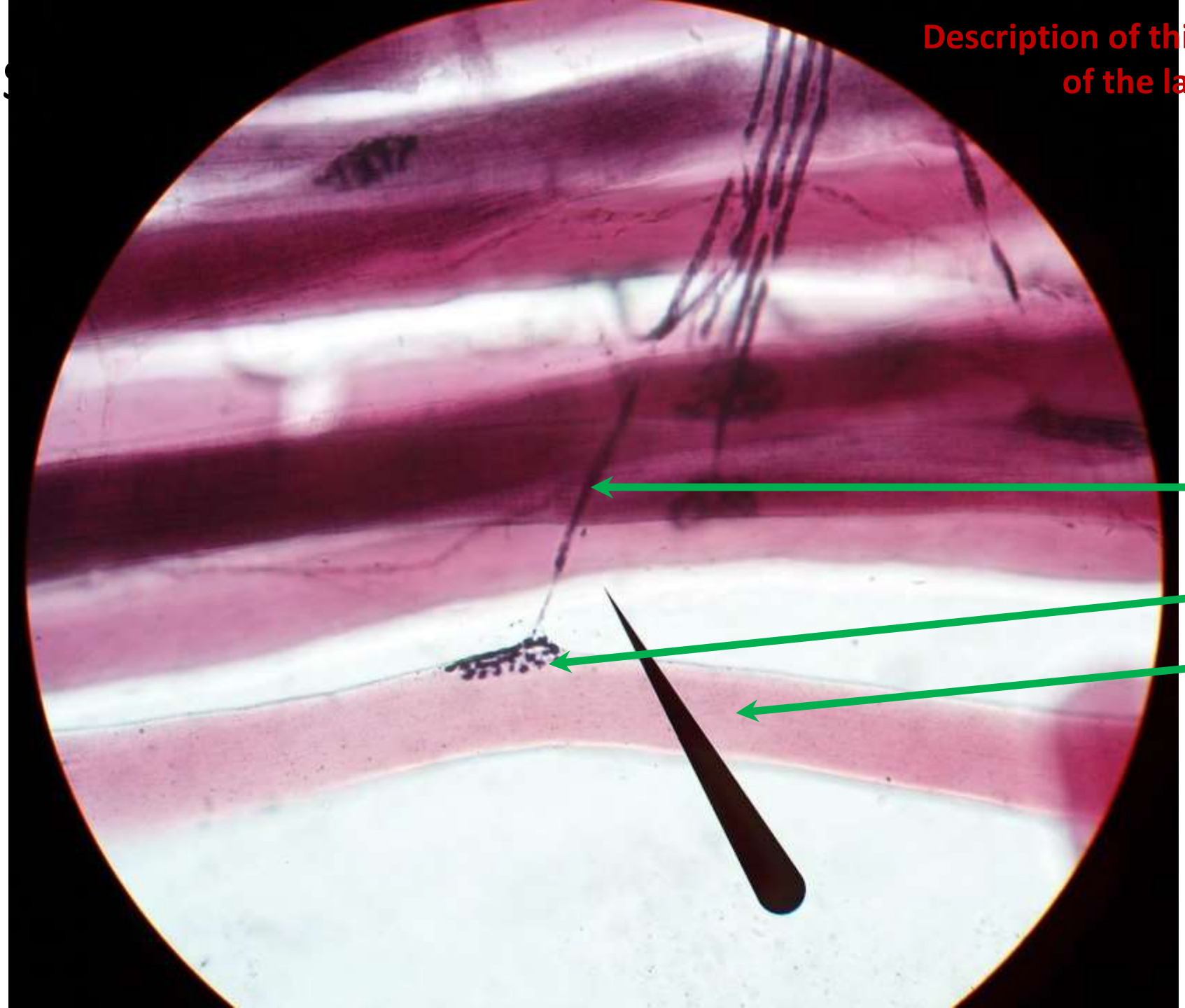

Description of this slide on page 9-3 of the lab manual